

Saksham Singh Kushwaha

Research Area: *Computer Vision, (Spatial) Audio, Multimodal*

Website: <https://sakshamsingh1.github.io/>

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EDUCATION

- **The University of Texas at Dallas** May 2027 (expected)
Ph.D. - Computer Science (Advisor: Dr. Yapeng Tian) GPA: 4.0/4.0
- **New York University, Courant** May 2023
M.S. - Computer Science GPA: 3.95/4.0
- **Indian Institute of Technology, Delhi** May 2018
B.Tech - Mathematics and Computing GPA: 7.97/10.0

EXPERIENCE

- **Sony Research – PhD Research Intern** Jan 2026 - Present
Topic: Spatial Audio-Video Models | Advisors: Kazuki Shimada, Takashi Shibuya, Koichi Saito Remote, USA
- **Adobe – PhD Research Intern** May 2025 - Aug 2025
Topic: Video object removal | Advisors: Kuldeep Kulkarni, Sayan Nag Bangalore, India
- **Dolby Laboratories – PhD Research Intern** May 2024 - Aug 2024
Topic: Spatial Audio Generation | Advisors: Avery Bruni, Jianbo Ma, Mark Thomas SF, USA
- **Nvidia – Deep Learning Intern** May 2022 - Aug 2022
Topic: Anomaly Detection | Advisors: Pradeep Thalasta, Arjun Chakraborty Remote, USA
- **Zomato – Machine Learning Engineer II** July 2018 - April 2021
Topic: Recommendation Systems | Advisors: Lalit Verma, Ram Singla Gurgaon, India

PUBLICATIONS

- **Object-WIPER: Training-free object and associated effect removal in videos (paper), (project page)**
Saksham Singh Kushwaha, Sayan Nag, Yapeng Tian, Kuldeep Kulkarni [CVPR 2026]
- **OmniSonic: Towards Universal and Holistic Audio Generation from Video and Text**
W. Pian, S. S. Kushwaha, Z. Chen, S. Deng, K. Wang, Y. Guo, Y. Tian [CVPR 2026]
- **AVRobustBench: Benchmarking the Robustness of Audio-Visual Recognition Models at Test-Time (paper)**
S. Maharana, S. S. Kushwaha, B. Zhang, A. Rodriguez, S. Wei, Y. Tian, Y. Guo [NeurIPS D&B 2025]
- **VinTAGe: Joint Video and Text Conditioning for Holistic Audio Generation (paper), (project page), (demo)**
Saksham S. Kushwaha, Yapeng Tian [CVPR 2025]
- **Diff-SAGe: End-to-end spatial audio generation using diffusion models (paper), (project page), (demo)**
Saksham S. Kushwaha, Jianbo Ma, Mark Thomas, Yapeng Tian, Avery Bruni [ICASSP 2025]
- **Audio-Visual Dataset Distillation (paper), (slides), (video)**
Saksham S. Kushwaha, Siva Vasireddy, Kai Wang, Yapeng Tian [TMLR 2024]
- **A multimodal prototypical approach for unsupervised sound classification (paper)**
Saksham S. Kushwaha, Magdalena Feuentes [INTERSPEECH 2023]
- **Sound source distance estimation in diverse and dynamic acoustic conditions (paper)**
Saksham S. Kushwaha, Iran Roman, Magdalena Feuentes, Juan Pablo Bello [WASPAA 2023]
- **Analyzing the effect of equal-angle spatial discretization on sound event localization & detection (paper)**
Saksham S. Kushwaha, Iran R. Roman, Juan Pablo Bello [DCASE 2022]

INVITED TALKS

- **Tutorial on creating webpage:** UT-Dallas, 2023 (slides)
- **Dataset distillation for Audio-visual tasks:** DL-MIR Workshop, Stanford University, 2024 (slides)
- **Object Removal in Videos:** Sony Research, 2026
- **Training-free Object Removal in Videos:** Voxel AI, 2026

ACADEMIC EXPERIENCE

- **Reviewer:** MLSP 2023, AAAI-24, ICASSP-24, NIPS-24, ACMMM-24, ICLR-25, ICML-25
- **Research Assistant:** Dr. Magdalena Fuentes (multimodal deep learning) [Aug'22-May'23], Dr. Raveesh Mayya (ML for digital policy change) [Aug'21-July'22], Dr. Yapeng Tian [Aug'24-Present]
- **Teaching Assistant:** Intro to python programming (Prof. Junpei Komiyama) [Jan'22-May'22], Discrete Mathematics (Prof. Simeon Ntafos) [May'23-Aug'23], Computer Science I (Prof. Scott Dollinger) [May'23-Aug'23]

STUDENT SUPERVISION

- **Preet Mukeshkumar Sojitra:** (MS Student at UT-Dallas)
- **Ahnaf Adib:** (BS at UT-Dallas)
- **Aditya Kulkarni:** (BS at UT-Dallas) → SWE @ Meta
- **Rushyam Keshri:** (MLE @ Zomato) → SWE III ML @ Google